

Unit-1

Introduction to Machine Learning



Prof. Jayesh D. vagadiya

Computer Engineering Department

Darshan Institute of Engineering & Technology, Rajkot

✉ jayesh.vagadiya@darshan.ac.in

☎ 9537133260



What is Machine Learning?

The Simple Definition

Machine Learning is like **teaching a computer to learn from experience**, just like a human! Instead of giving the **computer exact instructions for everything**, we give it lots of data, and it finds patterns to make **its own decisions**.

1. Virtual Assistants

Siri, Alexa, and Google Assistant use ML to recognize your voice and improve their understanding of your requests over time.

2. Spam Filters

Gmail learns to recognize "junk" emails by analyzing billions of messages and automatically moves them to your spam folder.

3. Recommendation Engines

Netflix and YouTube suggest videos based on your watch history, using ML to predict what you'll enjoy next.

4. Autonomous Vehicles

Self-driving cars use ML to detect objects, pedestrians, and traffic signs in real-time to navigate safely.

Machine Learning - Course Syllabus

Unit No.	Unit Title	Topics Covered	Hours	Weight %
1	Introduction to Machine Learning	Overview of AI, ML and DL, History of ML, How ML works, ML Life cycle, ML Terminology, Types of Learning: Supervised, Unsupervised, Reinforcement, Tools and Technology for Machine Learning, ML Application, issues in ML, Overview of Computational Learning Theory, Underfitting and Overfitting, Bias and Variance	8	20%
2	Regression	Introduction to Regression, Linear Regression, Multiple Linear Regression, Polynomial Regression, Ridge, Lasso, Logistic Regression, SVR, Evaluation techniques	8	15%
3	Classification & Ensembles	decision trees, probabilistic (conditional and generative), k- Nearest Neighbour, SVC Model Selection & Evaluation techniques: Ensemble Learning: Bagging, Random Forest, Boosting,	11	25%
4	Clustering & Dim. Reduction	Distance based clustering, Hierarchical Clustering, Density based clustering, Outlier Detection Methods, Overview of Dimensionality Reduction: Feature Selection techniques : Filter, Wrapper, Embedded, Feature Extraction techniques, Dimensionality Reduction techniques: Principal Component Analysis, t-SNE	10	20%
5	Artificial Neural Networks	Overview of Artificial Neural Networks, History of ANN, How ANN works Multilayer Perceptron, Activation Functions and Hyper parameter Tuning, Back propagation, multilayer neural network, Regularization, Introduction of CNN, RNN.	8	20%

Key Libraries for Machine Learning

NumPy

Numerical Python

The fundamental package for scientific computing with Python.

- Supports large, multi-dimensional arrays and matrices.
- High-level mathematical functions for array operations.
- Vectorized computation for efficiency.

Pandas

Data Manipulation

High-performance, easy-to-use data structures and data analysis tools.

- DataFrame objects for tabular data.
- Tools for reading/writing data in text/CSV formats.
- Data alignment, reindexing, and filtering.

Scikit-Learn

ML Algorithms

Simple and efficient tools for predictive data analysis.

- Implements Supervised & Unsupervised learning.
- Regression, Classification, & Clustering.
- Model selection and evaluation techniques.

PyTorch

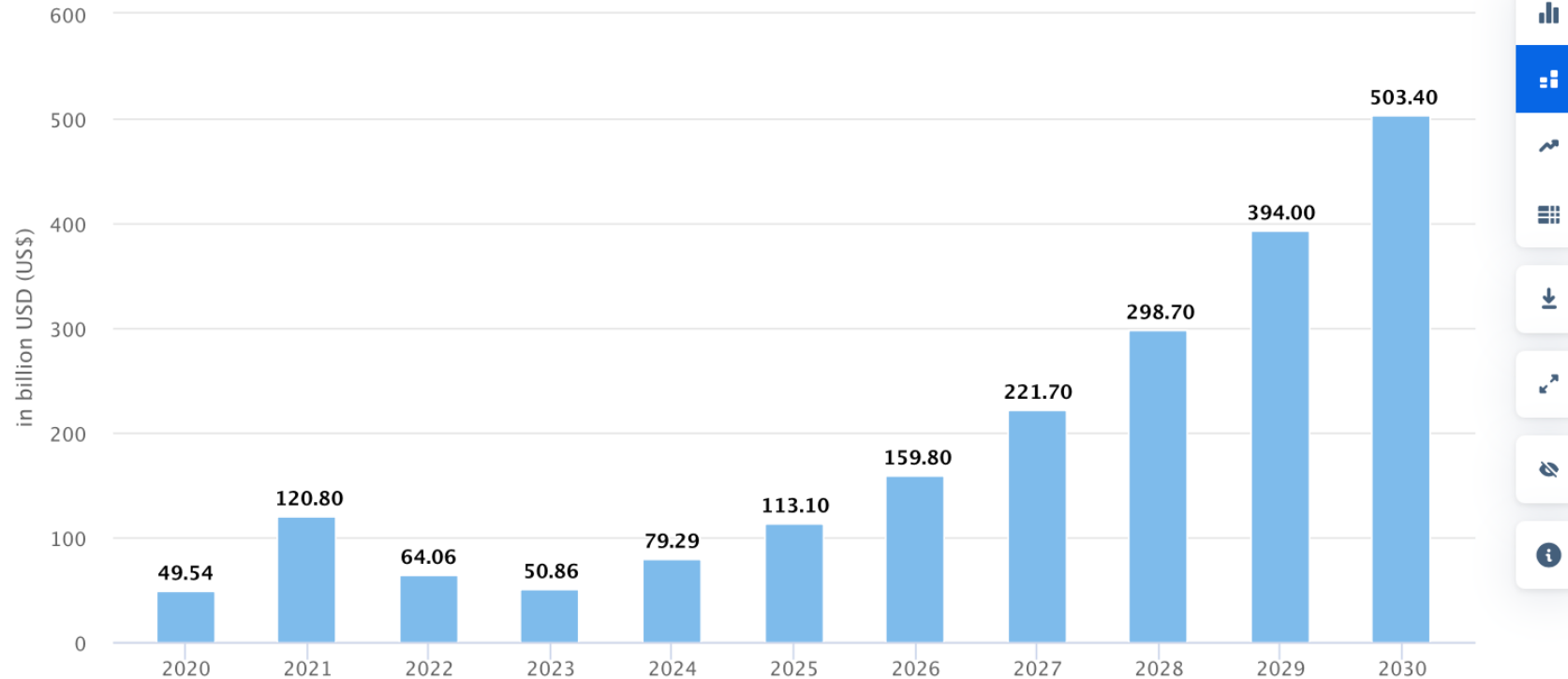
Deep Learning

An open-source machine learning framework based on the Torch library.

- Used for Artificial Neural Networks (ANN).
- Dynamic computational graphs.
- Highly efficient for Computer Vision and NLP.

ML – Market size

MARKET SIZE



Notes: Data was converted from local currencies using average exchange rates of the respective year.

Most recent update: Mar 2024

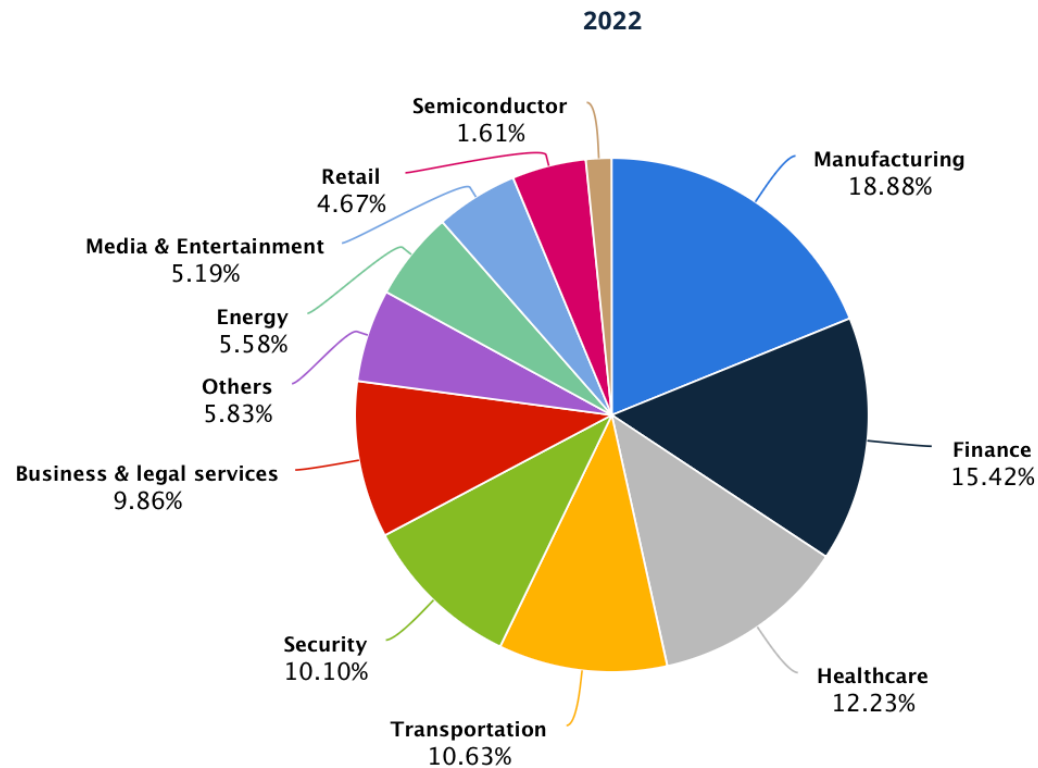
Source: Statista Market Insights

Ref: <https://www.statista.com/outlook/tmo/artificial-intelligence/machine-learning/worldwide#global-comparison>

ML – Market size share by industry

MARKET SIZE SHARE BY INDUSTRY

i in percent



Most recent update: Mar 2024

Source: Statista Market Insights

Ref: <https://www.statista.com/outlook/tmo/artificial-intelligence/machine-learning/worldwide#global-comparison>

Machine Learning - Statistic/Insight

Category	Statistic/Insight
Adoption	48% of businesses globally use machine learning.
Use Cases	- 57% of companies use machine learning to improve the consumer experience. - 49% use it for marketing and sales.
Market Growth	The machine learning market is expected to grow at a CAGR of 38.8% from 2022 to 2029.
Employment	Employment opportunities for machine learning engineers are expected to grow at a rate of 22% globally from 2020 to 2030.
AI Adoption	91.5% of top companies invest in artificial intelligence.
AI in Financial Services	IBM identifies financial services as the leading sector for AI adoption.
Process Automation	Process automation is the most common application for AI and machine learning.

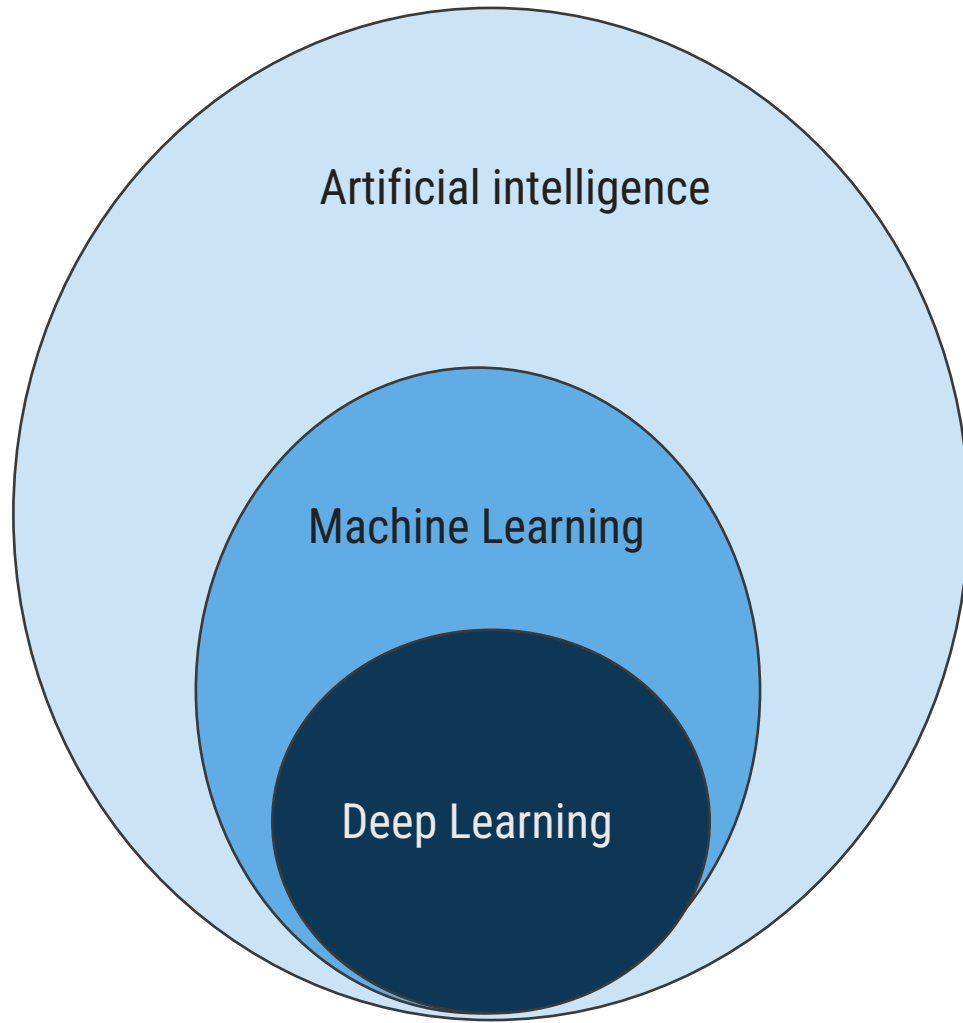
Ref: <https://learn.g2.com/machine-learning-statistics#:~:text=57%25%20of%20companies%20and%20businesses,businesses%20globally%20use%20machine%20learning.>



Topics to be covered

- Motivation
- Machine Learning - Statistic/Insight
- Machine Learning - Big Picture
- Traditional Programing Vs Machine Learning
- Why Use Machine Learning?
- When to Use Machine Learning ?
- ML Terminology
- Machine Learning Project life cycle
- Classification of ML Techniques
- Type of Machine Learning Algorithms
- Main Challenges of Machine Learning

Big Picture



Artificial intelligence

A technique which enables **machines to mimic human** behaviour

Machine Learning

Subset of AI technique which use **statistical methods to enable machines to improve** with experience

Deep Learning

Subset of ML **which make the computation of multi-layer neural network feasible**

History of ML

▶ 1950s: Foundations

- Alan Turing: **Turing Test** (1950) – Can machines think?
- Arthur Samuel: Coined "**Machine Learning**" and **created a Samuel's Checkers Player(1959)**.
- Perceptron (1958): Frank Rosenblatt developed the **first neural network model**.

▶ 1960s-1970s: Rise of Symbolic AI

- Expert **Systems dominated** (e.g., DENDRAL - how scientists form and discover hypotheses.).
- Limited **progress due to computational constraints and lack of large datasets**.

▶ 1980s: Connectionism Revival

- Introduction of Backpropagation (1986): **A breakthrough in training neural networks**.
- Neural networks **gained traction for pattern recognition tasks**.

History of ML

▶ 1990s: Statistical Learning

- Shift from rule-based systems to statistical approaches.
- Development of SVMs (Support Vector Machines) and Ensemble Methods.

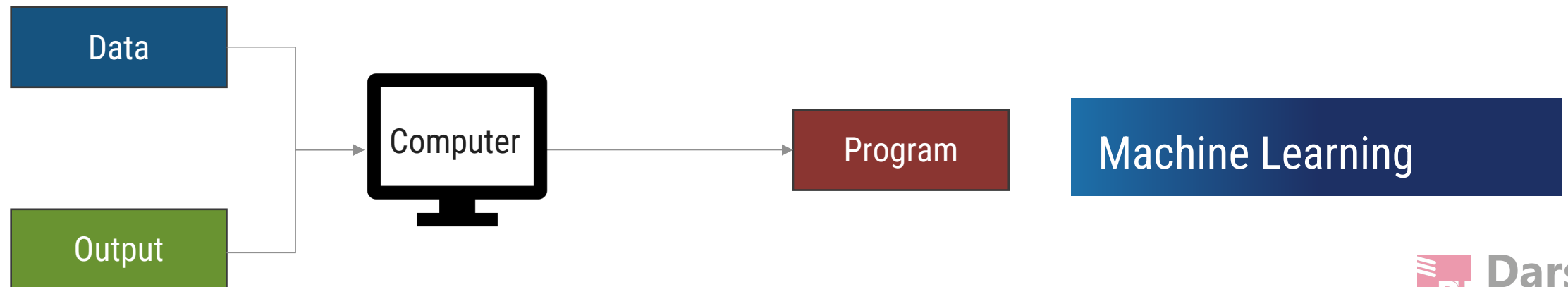
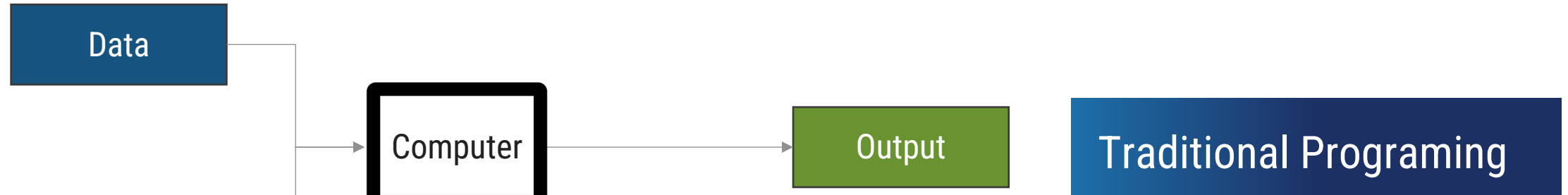
▶ 2000s: Big Data and GPUs

- Growth of large datasets (e.g., ImageNet).
- GPUs accelerated deep learning computation.

▶ 2010s: Deep Learning Revolution

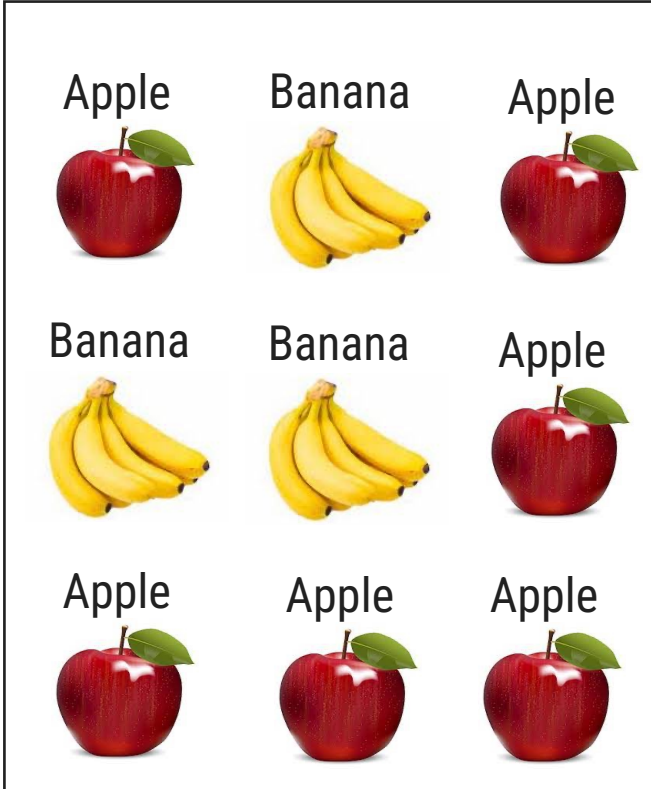
- Breakthroughs in computer vision (AlexNet, 2012) and NLP (transformers).
- Applications expanded to healthcare, autonomous vehicles, and more.

Traditional Programming Vs Machine Learning



Machine learning

Input Data



Input Data Set
for Learning



???

Algorithm

Prediction

Its an
Apple

Definition

Tom M. Mitchell
(1997)

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E

Arthur Samuel
(1959)

Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.

Ian Goodfellow, Yoshua
Bengio, and Aaron
Courville (2016):

Machine learning is the science of getting computers to act without being explicitly programmed, by extracting patterns and insights from data

Ethem Alpaydin
(2004):

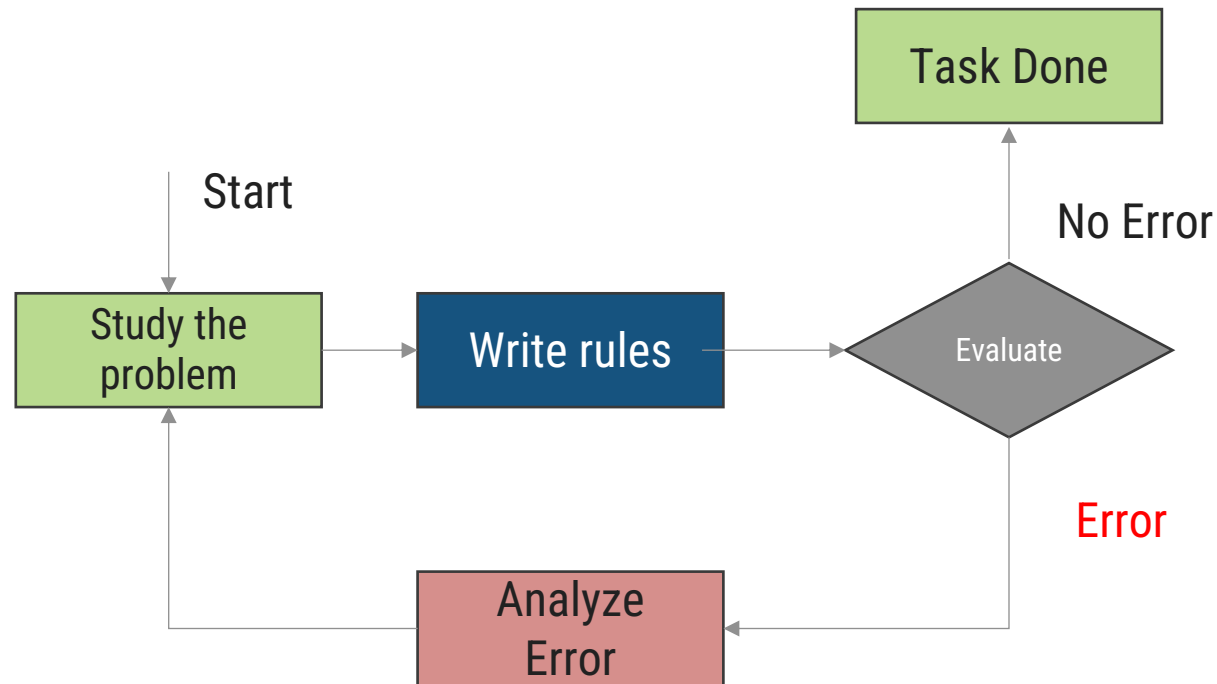
Machine learning develops algorithms that can learn from and make predictions on data.

Kevin P. Murphy
(2012)

Machine learning is a set of methods that can automatically detect patterns in data and then use the uncovered patterns to predict future data or other outcomes of interest

Why Use Machine Learning?

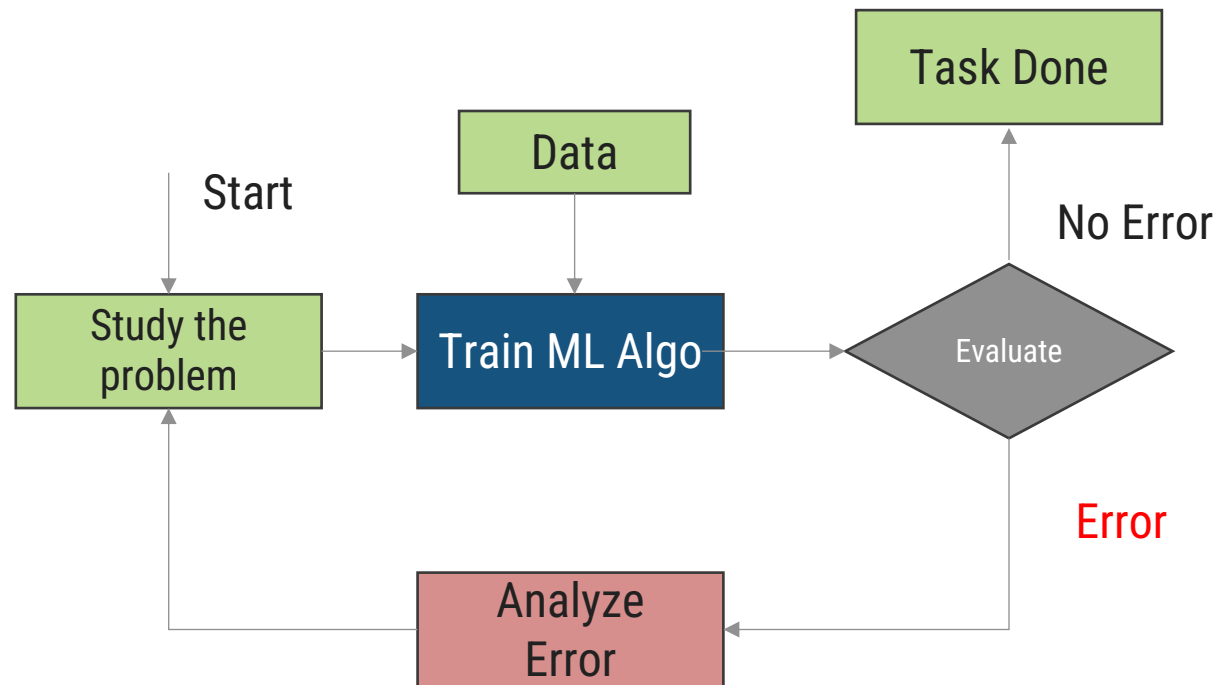
SPAM FILTER -> Traditional Programming Techniques.



- ▶ **Hard-coded** rules
- ▶ Rules **are static**, requires manual updates.
- ▶ Moderate, prone to **errors**
- ▶ High **Maintenance** of rules.
- ▶ Poor for **large datasets**

Why Use Machine Learning?

SPAM FILTER -> The Machine Learning approach



- ▶ Data-driven patterns
- ▶ Dynamic, learns from data
- ▶ Accuracy High, especially with quality data.
- ▶ Lower Maintenance (after initial setup)
- ▶ Scales well with more data

When to Use Machine Learning ?

When the Problem Is Too Complex for Coding

- ▶ In a situation where the problem is **so complex or big that** you cannot hope to write **all the rules to solve it** and **where a partial solution is viable** and interesting, you can try to solve the problem with machine learning.

When the Problem Is Constantly Changing

- ▶ Some **problems may continuously change with time so that the programming** code must be regularly updated.
- ▶ That results in the frustration of software engineers working on the problem, an increased chance of **introducing errors, difficulties of combining** “previous” and “new” logic, and significant overhead of testing and **deploying updated solutions**.

When It Is a Perceptive Problem

- ▶ Today, it's hard to **imagine someone trying to solve perceptive problems** such as speech, **image, and video** recognition without using machine learning

When to Use Machine Learning

When It Is an Unstudied Phenomenon

- ▶ If we need to be able to make **predictions of some phenomenon** that is not well-studied **scientifically**, but examples of it are **observable**, then machine learning might be an appropriate.

When It Is Cost-Effective

- ▶ The cost of training the model includes **human labor and, in some cases, the expensive hardware** needed to train deep models.
- ▶ Model maintenance includes **continuously monitoring** the model and **collecting additional data** to keep the model up to date

When Not to Use Machine Learning

- ▶ Every action of the system or a **decision made by it must be explainable**.
- ▶ Every change in the **system's behavior compared** to its past behavior in a similar situation must be **explainable**.
- ▶ The cost of an **error** made by **the system** is too high.
- ▶ Getting the right data is **too hard or impossible**.
- ▶ You can solve the problem using **traditional software development** at a **lower cost**.
- ▶ A simple **heuristic would** work **reasonably well**.
- ▶ You build a **system that will not have to be improved** frequently over time.

ML Terminology



Algorithm

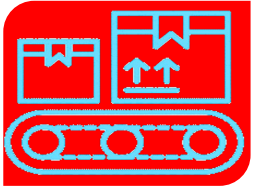
- ▶ A method, function, or series of instructions used to generate a machine learning model.
- ▶ **Examples:** include linear regression, decision trees, support vector machines, and Naïve Bayes.



Model

- ▶ A mathematical representation of a real-world process learned from data.
- ▶ A model defines the relationship between features and label.
- ▶ Models are created/learned when you train an algorithm on a dataset.
- ▶ **Example:** A spam filter

ML Terminology



Feature

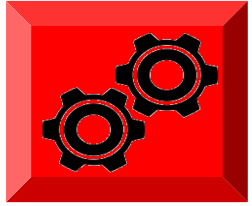
- ▶ With respect to a dataset, a feature **represents an attribute** and value combination.
- ▶ A Feature is the input **used for prediction or classification**.
- ▶ Input variables for the model
- ▶ **Example :** "email length" and "presence of keywords"



Label

- ▶ In supervised learning, a **label is the "answer" or "result"** portion of an Model.
- ▶ **Target outcomes.**
- ▶ **Example:** "Spam" or "Not Spam"

ML Terminology



Training

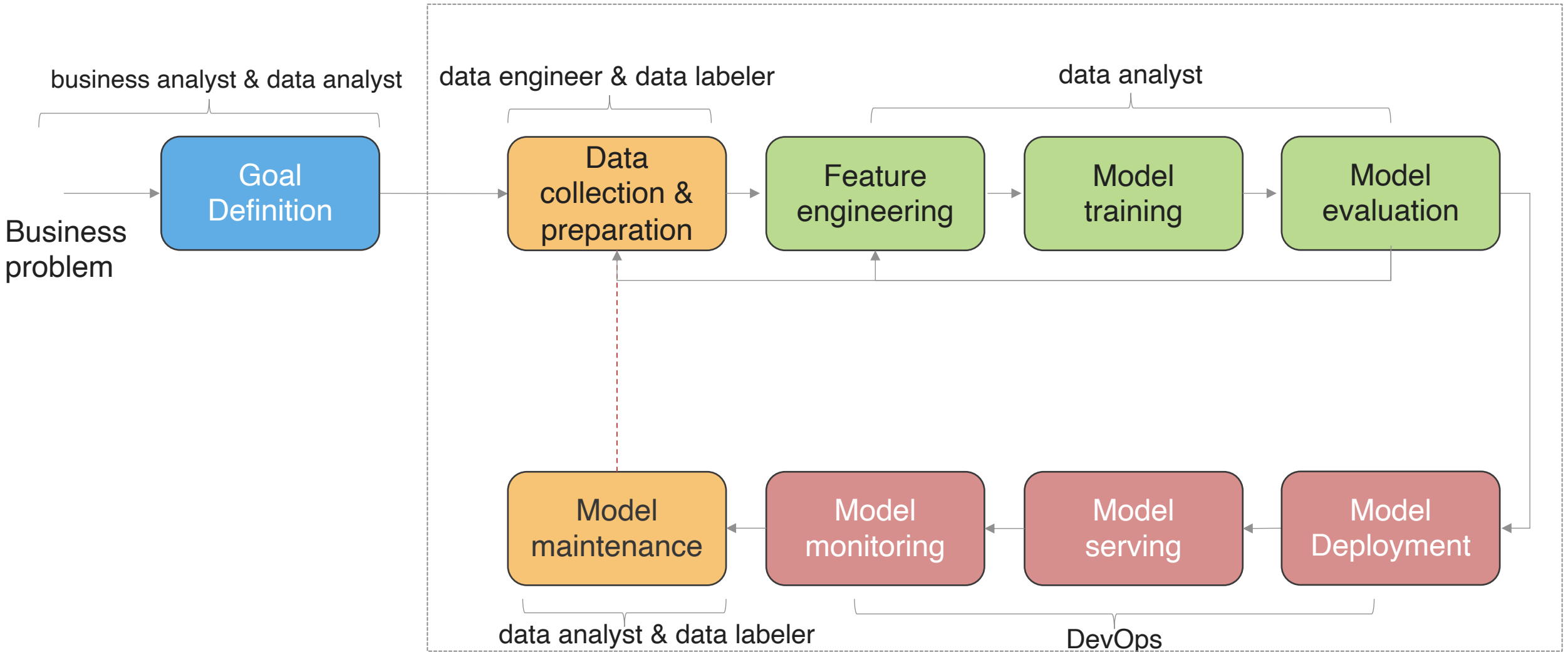
- ▶ Data used to **teach** the model .
- ▶ Training means **creating or learning** the model.
- ▶ During training the model **is given labeled examples**.
- ▶ Training enables the model to gradually **learn the relationships between features and label**.
- ▶ **Example:** set of emails labeled as spam or not for training.



Testing

- ▶ Data used to **evaluate** the model.
- ▶ Testing means **applying the trained model to unlabeled** examples.
- ▶ That is, you use the trained model to make useful predictions.
- ▶ **Example:** unseen emails

Machine Learning Project life cycle



Machine Learning

Machine Learning Project life cycle

► Problem Definition

- Define the **business or research problem**.
- Identify **objectives, success metrics, and constraints**.
- Understand the potential **impact of ML on solving the problem**.

► Data Collection & Preparation

- Gather **relevant data from various sources** (databases, APIs, web scraping, etc.).
- Ensure **data aligns with the problem's needs** in terms of scope, quality, and scale.
- Clean the data: **Handle missing values, remove duplicates, and address** inconsistencies.
- Normalize or **standardize features** to ensure compatibility with ML algorithms.
- Split the **dataset into training, validation, and testing subsets**.
- Perform exploratory data analysis (**EDA**) to understand data distributions and patterns.

► Feature Engineering

- Select or create meaningful **features from raw data to improve model performance**.
- Techniques include: **Feature selection , Feature transformation , Feature creation**

Machine Learning Project life cycle

► Model Training

- Train the model **using the training dataset**.
- Optimize model **parameters using techniques like gradient descent**.
- Monitor performance using metrics **like accuracy, precision, recall**, etc.

► Model Evaluation

- Evaluate the model on the **validation dataset to tune hyperparameters**.
- Assess performance using appropriate **metrics accuracy, precision, recall**, etc.
- Perform **cross-validation for robust evaluation**.

► Model Deployment

- Integrate the **trained model into production environments**.
- Options for deployment:
 - **Batch processing** (offline).
 - **Real-time inference** (APIs, cloud services).

Machine Learning Project life cycle

► Model Serving

- Ensures the **model is operational** and **accessible to users**.
- Serving in **Batch Mode** - In **batch serving**, predictions are generated for a large dataset all at once and stored for later use.
- In on-demand serving, **predictions are generated dynamically** in response to a specific user request.

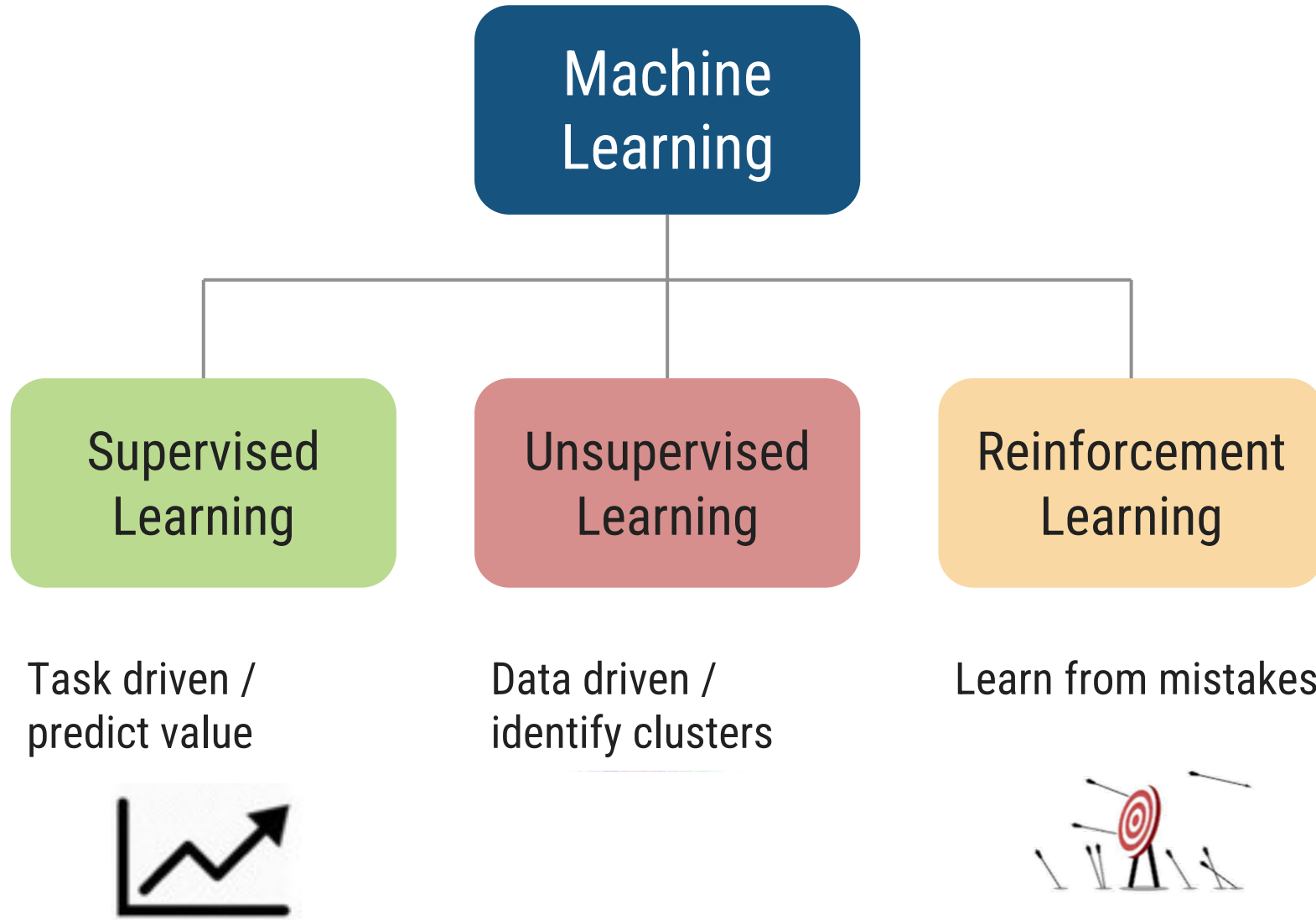
► Model Monitoring

- Model **Monitoring detects performance issues** or **changing data distributions**.
- The process of tracking a deployed model's performance, accuracy, and reliability in production.

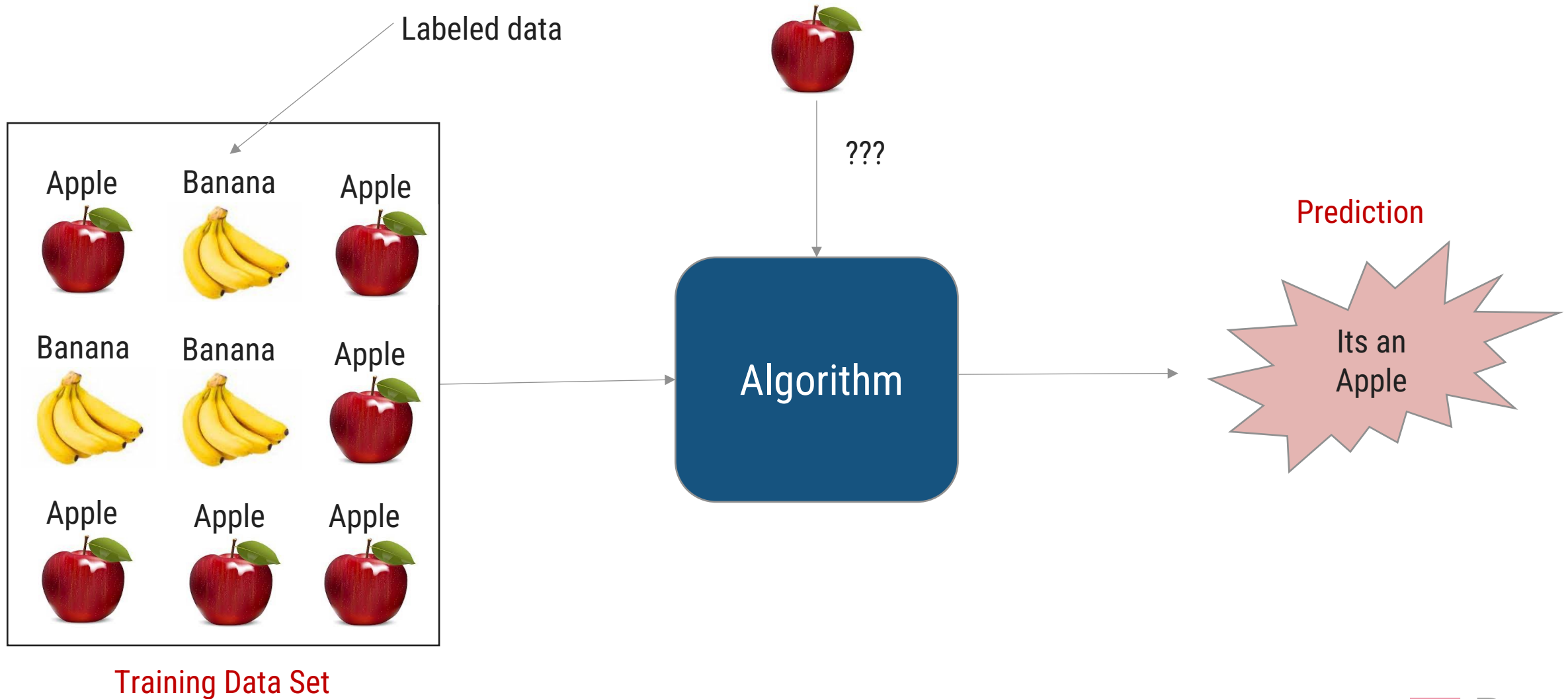
► Model Maintenance

- Model Maintenance **addresses issues found during monitoring** and **improves the model's accuracy** and reliability.
- Activities aimed at ensuring the model remains **accurate, relevant, and efficient over time**.

Classification of ML Techniques

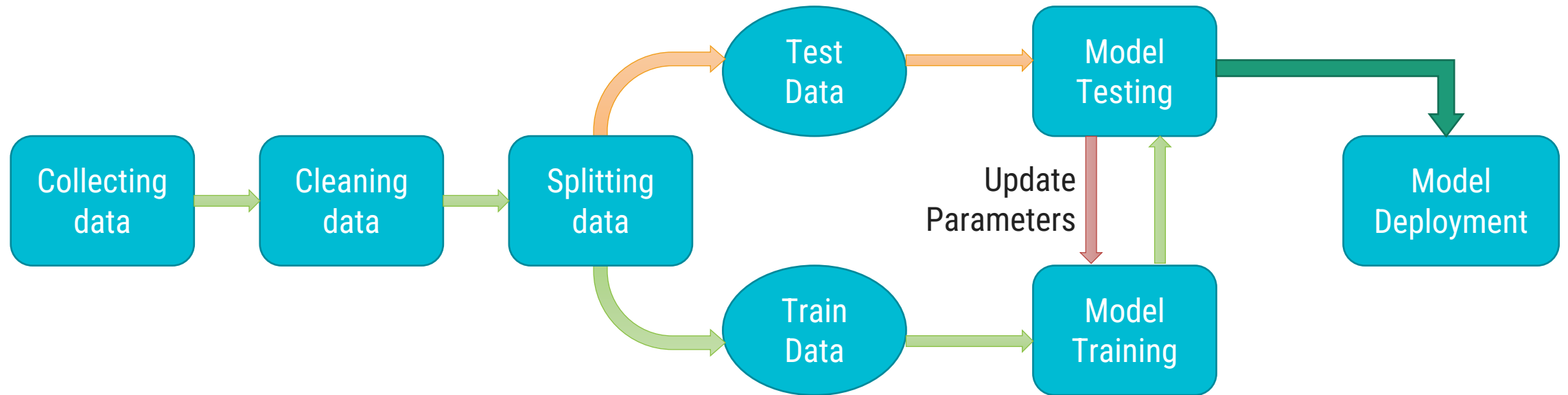


Supervised Learning

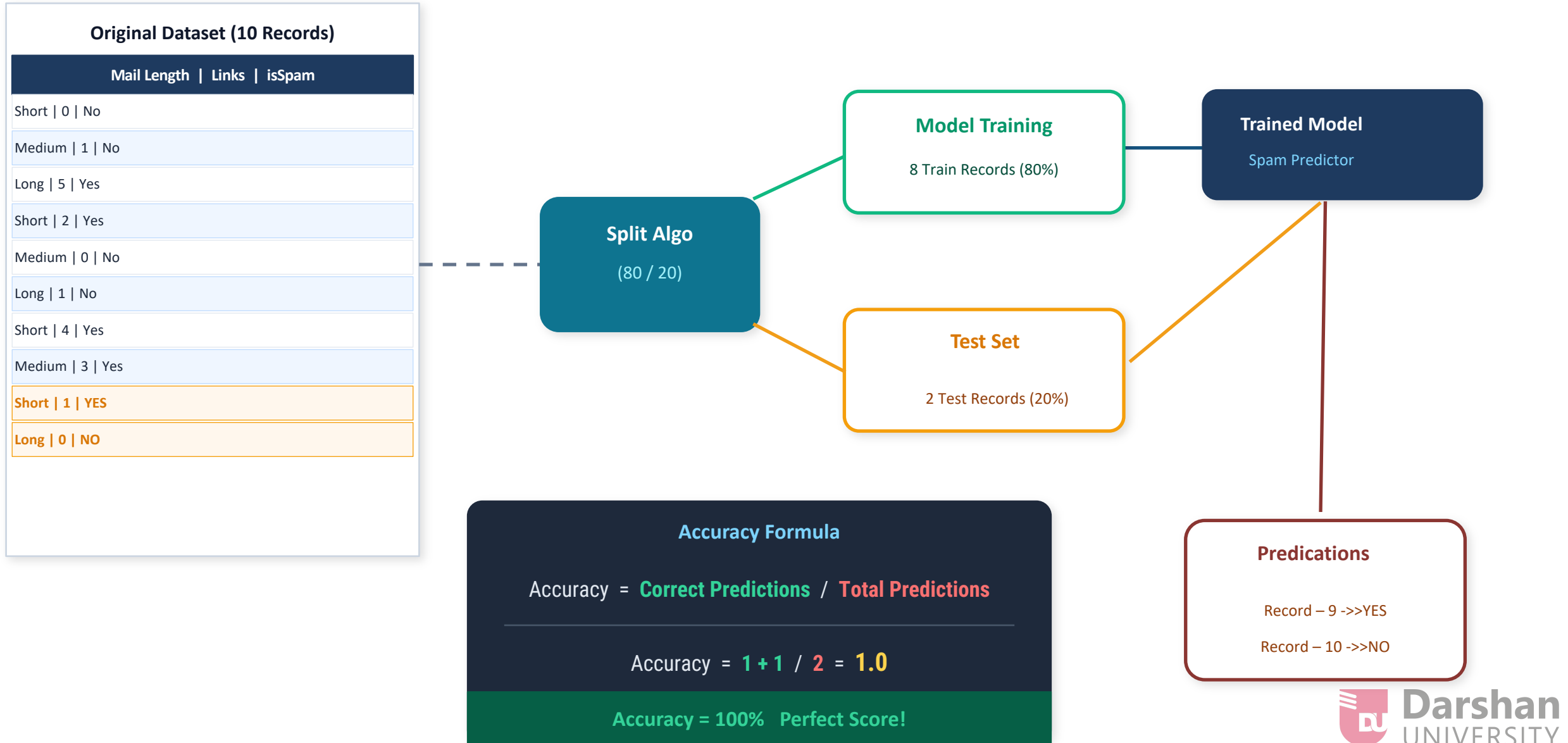


Here we Use Label Data to train Classifier (Supervised Learning)

Machine Learning Process for Supervised Learning



Machine Learning Process for Supervised Learning



Supervised Learning

- ▶ Also called **predictive learning**. A machine predicts the class of unknown objects based on prior **class-related information of similar objects**.
- ▶ The major motivation of **supervised learning is to learn from past information**.
- ▶ A machine needs the basic information to be provided to it. This basic input, or the experience in the paradigm of machine learning, is **given in the form of training data**.
- ▶ Example:
 - Predicting the results of a game
 - Predicting whether a tumour is malignant or benign
 - Predicting the price of domains like real estate, stocks, etc.
 - Classifying texts such as classifying a set of emails as spam or non-spam

Supervised Learning

- ▶ in **predictive (supervised) learning**, the goal is to approximate the true function $f(x)$, which maps input x to output y .
- ▶ This is achieved **by learning a hypothesis function $h(x)$** that is as close as possible to $f(x)$. Here's how this works:

1. What we have (The Given)

A Training Dataset:

$$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

x: The input features (like area of house)

y: The correct answer (like price of house)

We know y comes from some hidden rule $y = f(x)$ that we are trying to figure out.

2. What we want (The Goal)

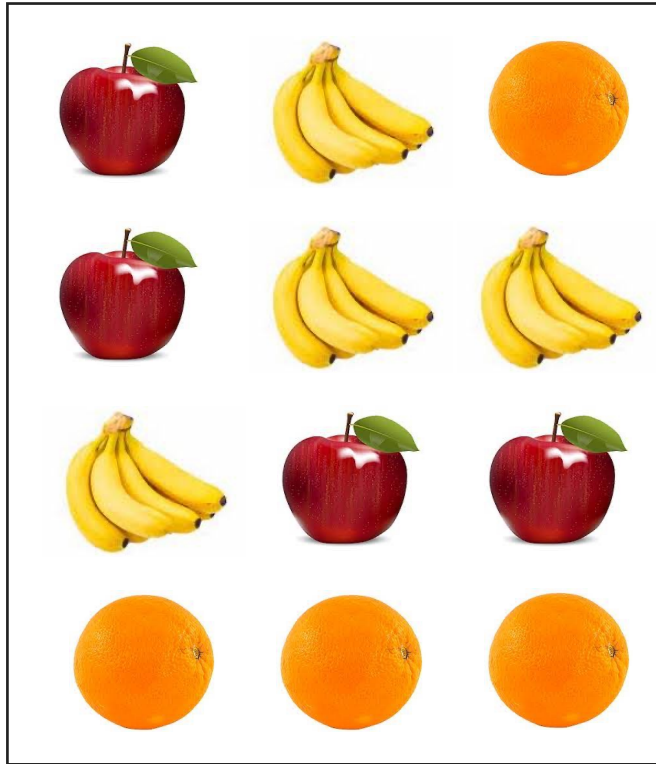
Create a Model (h):

$$h(x) \approx f(x)$$

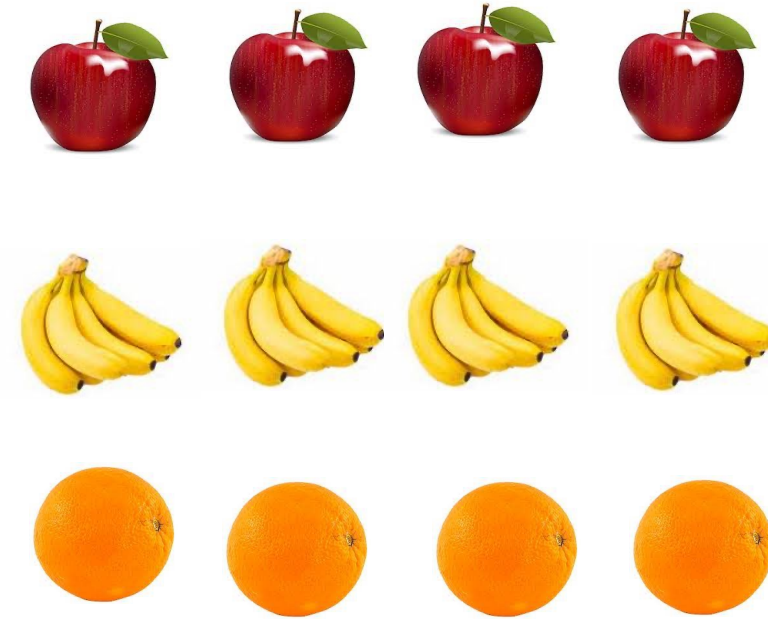
- Train the model $h(x)$ to match the real rule $f(x)$ as closely as possible.
- The model should predict y accurately even for **new data** it hasn't seen before!

Unsupervised Learning

Input Data



Clustering
Algorithm



Here we don't use Label Data to train Classifier(Unsupervised Learning)

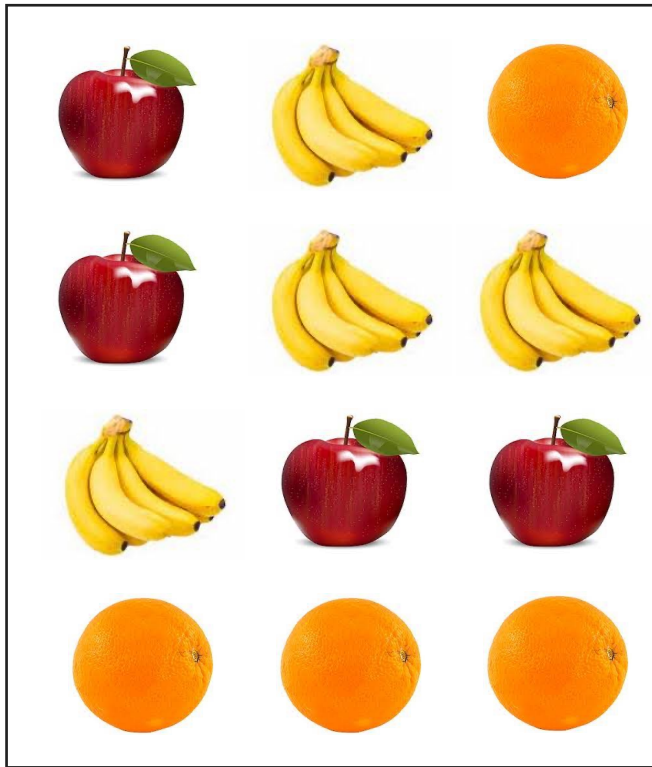
Finding **similarities** between data according to the **characteristics found in the data** and grouping similar data **objects** into **clusters**

Unsupervised Learning

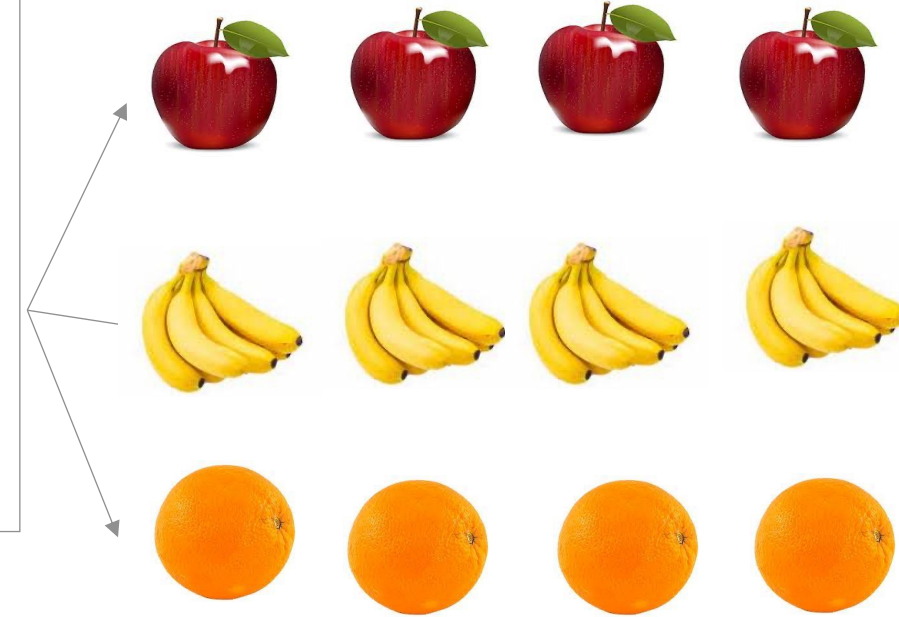
- ▶ Also called **descriptive learning**. A machine finds **patterns** in unknown objects by **grouping similar objects together**.
- ▶ In unsupervised learning, the **objective is to take a dataset as input** and try to find natural groupings or patterns within the data elements or records.
- ▶ Therefore, **unsupervised learning is often termed as descriptive model** and the process of unsupervised learning is referred as pattern discovery or knowledge discovery.
- ▶ One critical application of **unsupervised learning is customer segmentation**.
- ▶ **Clustering** is the main type of unsupervised learning. It intends to group or organize similar objects together.
- ▶ For that reason, **objects belonging to the same cluster are quite similar** to each other while objects belonging to **different clusters are quite dissimilar**.
- ▶ Clustering, Dimensionality Reduction, Density Estimation, Association Rule Learning are the type of unsupervised learning.

Reinforcement Learning

Input Data



Reinforcement Learning Model

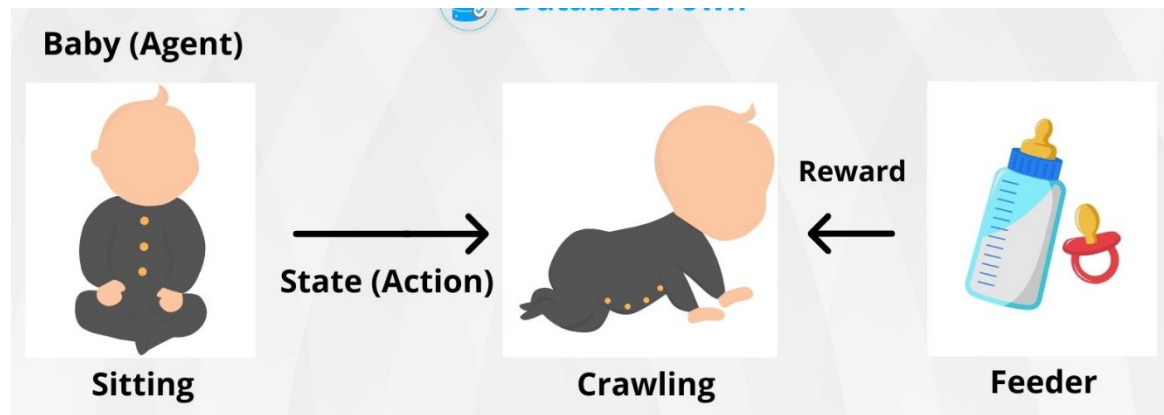


Reinforcement Learning

- ▶ It use **feedback** is called a **reward**, or reinforcement.
- ▶ 1) The **agent**(decision-maker) **observes the current state** of the environment(The system or scenario with which the agent interacts.).
- ▶ 2) **Based on the state**, the agent **chooses an action according to its policy**(The agent's strategy, mapping states to actions.).
- ▶ 3) The **environment responds to the action** by:
 - ↳ Transitioning to a **new state**
 - ↳ Providing a **reward**.
- ▶ 4) The agent **updates its policy** using the feedback.
- ▶ Steps 1–4 repeat, with the goal of **maximizing the total reward over time**.
- ▶ Reinforcement learning is like **training a dog**, where the agent (dog) learns to take actions **based on rewards from interacting with its environment**.
- ▶ The key components include the **agent (dog)**, **environment** (surroundings), **state** (current situation), **action** (behavior), and **reward** (feedback).

Reinforcement Learning

- ▶ The dog **learns through trial and error**, associating positive **rewards like treats** with **correct actions** and negative rewards with incorrect ones.
- ▶ Initially, the **dog explores various actions to identify what yields rewards**, then exploits this knowledge to **consistently perform rewarding actions**.
- ▶ Concepts like **positive reinforcement** (rewarding good behavior) and shaping (rewarding gradual improvements) are **central to both dog training and reinforcement learning**.

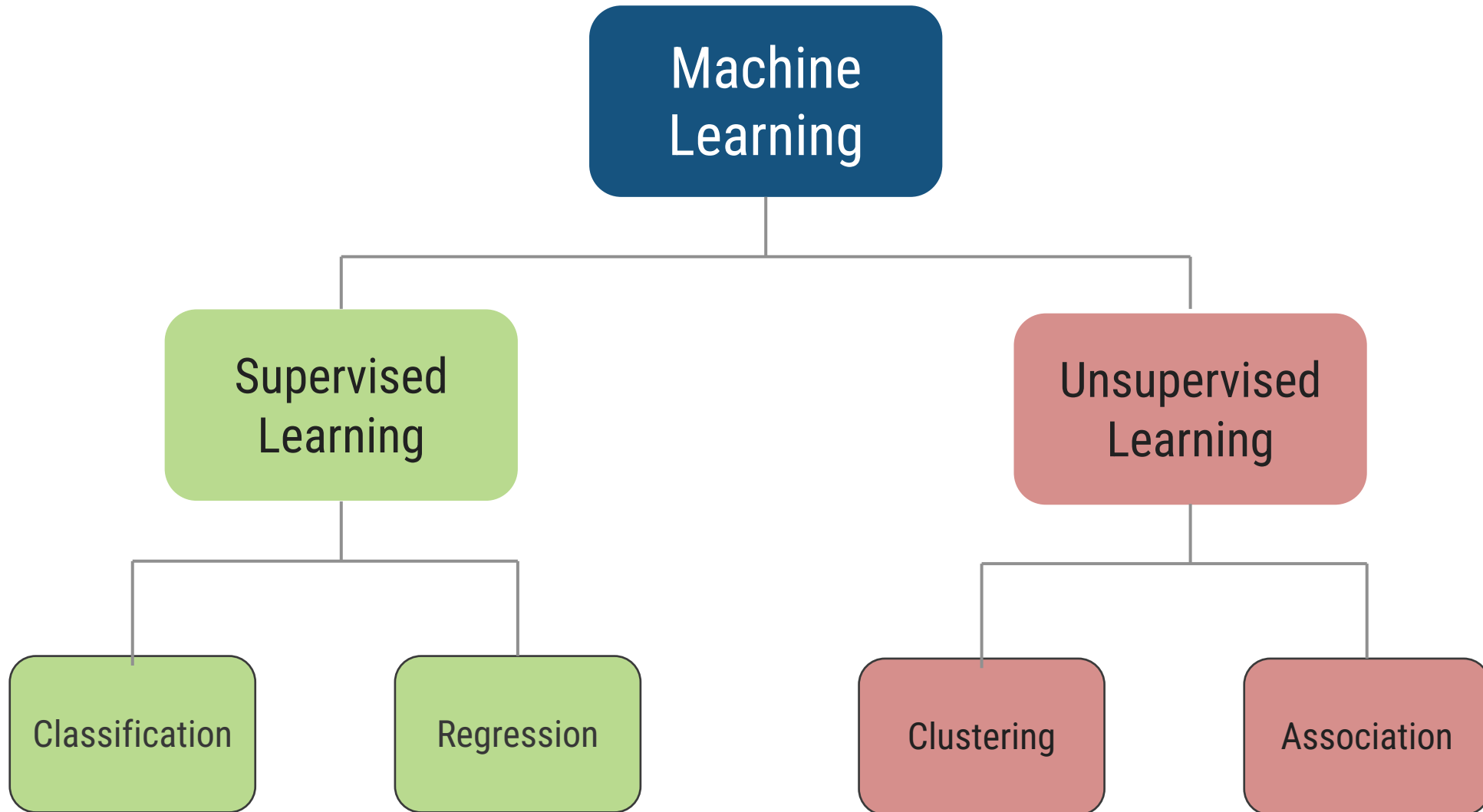


Ref: <https://databasetown.com/basics-of-reinforcement-learning/>

Supervised vs. Unsupervised vs. Reinforcement

Supervised Learning	Unsupervised Learning	Reinforcement
Input data is labeled	Input data is unlabeled	Sequential interaction data
Has a feedback mechanism	Has no feedback mechanism	Reward signals
Data is classified based on the training dataset	Assigns properties of given data to classify it	The model learn and updates itself through reward.
Divided into Regression & Classification	Divided into Clustering & Association	No Types
Used for prediction	Used for analysis	Used for Maximize rewards
Algorithms include: decision trees, logistic regressions, support vector machine	Algorithms include: k-means clustering, hierarchical clustering, apriori algorithm	Algorithms include: Q-Learning, Sarsa
A known number of classes	A unknown number of classes	A unknown number of classes

Type of Machine Learning Algorithms



Classification and Regression

Regression

What will be the **next increment**?

3.9%



Employee

Classification

Is the employee "**eligible for promotion**" or "**not eligible**" ?

Eligible



Employee

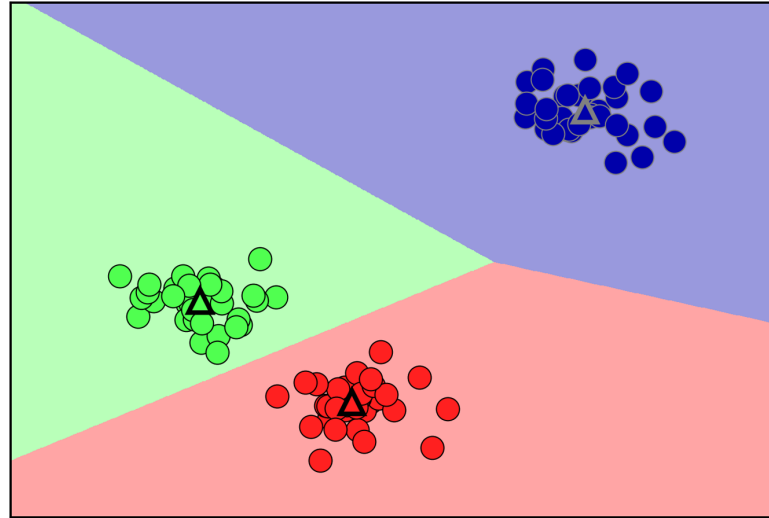
Classification and Regression

- ▶ **Regression** is used when you want to predict a numerical value (e.g., predicting salary, sales, temperature).
- ▶ **Classification** is used when you want to assign a category label to data (e.g., classifying emails as spam or not, identifying diseases, or segmenting customers).

Regression	Classification
- Predict Continuous numerical value (real number)	- Predict Discrete class label (category)
- Predicting house price - Forecasting, trend analysis, estimating quantities	- Spam email detection - Categorizing data, identifying patterns, classification tasks
quantitative	qualitative

Clustering

- ▶ Cluster is a **group of similar data points**.
- ▶ Points in the **same cluster are more similar** to each other than to those in other clusters.



- ▶ Clustering **often uses distance measures to group points**.
- ▶ Common metrics: **Euclidean Distance, Manhattan Distance, Cosine Similarity**

Clustering

- ▶ Minimize intra-cluster distance (points within the same cluster are close).
- ▶ Maximize inter-cluster distance (clusters are distinct).
- ▶ **Cluster analysis** or **simply clustering** is the process of **partitioning a set of data objects** (or observations) into **subsets**.
- ▶ Each subset is a cluster.
- ▶ The set of clusters resulting from a cluster analysis can be referred to as a clustering.
- ▶ **Cluster analysis (or *clustering, data segmentation, ...*)**
 - ➔ Finding **similarities between data** according to the **characteristics found in the data** and grouping similar data objects into clusters
- ▶ Clustering is useful in that it can lead to the discovery of **previously unknown groups** within the data.

Association Rule Mining

- ▶ Given a set of **transactions**, we need **rules** that will predict the occurrence of an item based on the occurrences of other **items in the transaction**.
- ▶ **Market-Basket transactions**

TI D	Items
1	Bread, Milk
2	Bread, Chocolate, Pepsi, Eggs
3	Milk, Chocolate, Pepsi, Coke
4	Bread, Milk, Chocolate, Pepsi
5	Bread, Milk, Chocolate, Coke

Example of Association Rules

$\{\text{Chocolate}\} \rightarrow \{\text{Pepsi}\},$
 $\{\text{Milk, Bread}\} \rightarrow \{\text{Eggs, Coke}\},$
 $\{\text{Pepsi, Bread}\} \rightarrow \{\text{Milk}\}$

Market Basket Analysis

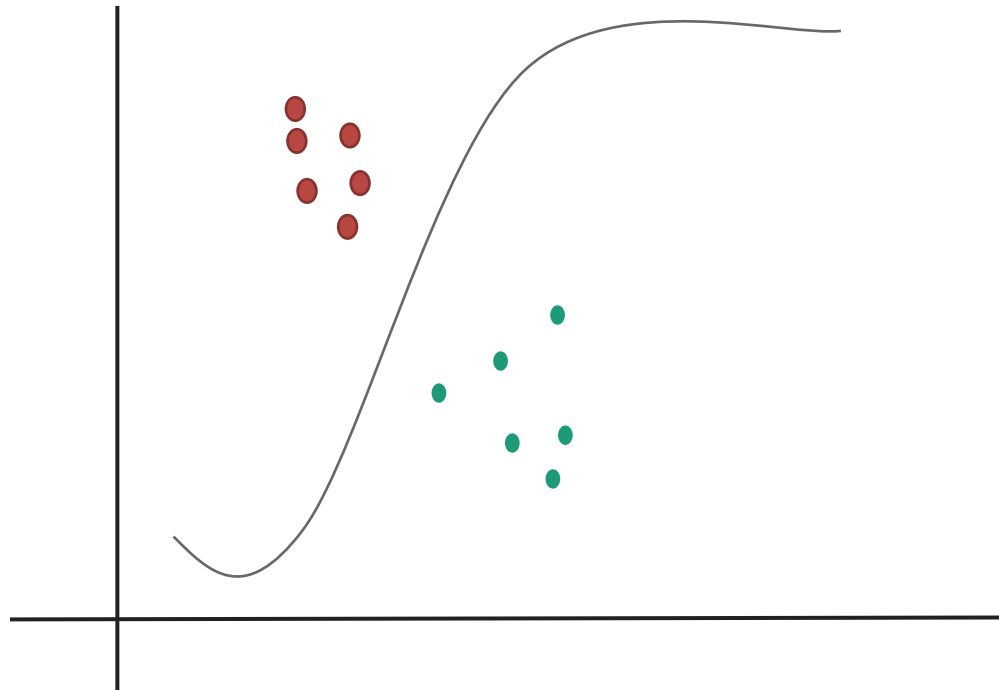


- ▶ Market Basket Analysis is a **modelling technique to find frequent itemset**.
 - ▶ It is based on, if you buy a certain group of items, you are more (or less) likely to buy another group of items.
 - ▶ For example, if you are in a store and you buy a car then you are more likely to buy insurance at the same time than somebody who don't buy insurance also.
 - ▶ The **set of items** that a customer buys it referred as an **itemset**.
 - ▶ Market basket analysis seeks to **find relationships between purchases** (Items).
 - ▶ **E.g.** IF {Car, Accessories} THEN {Insurance}
- {Car, Accessories} → {Insurance}

Model-Based vs. Instance-Based Learning

Model-Based

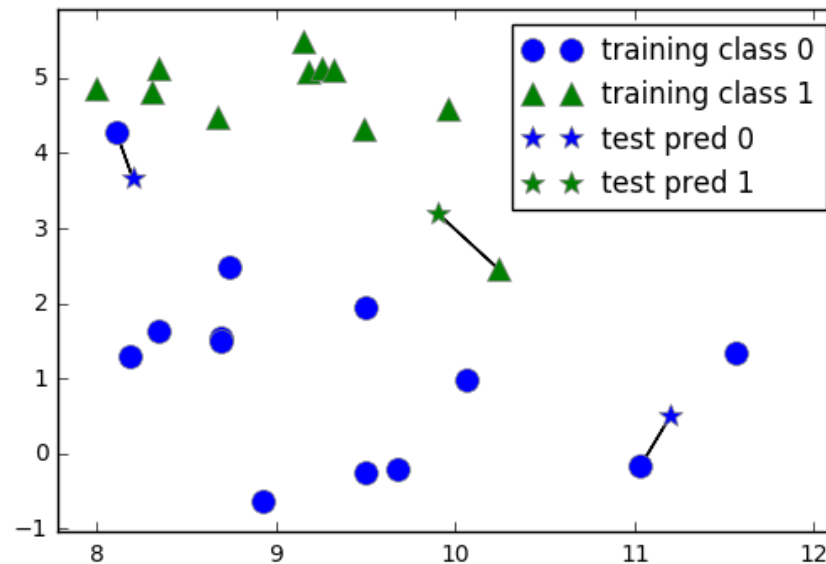
- ▶ Most **supervised** learning algorithms are model-based.
- ▶ Model-based learning algorithms use the training data **to create a model** with parameters learned from the **training** data.
- ▶ After the model is trained, it can be **saved on disk while** the training data can be **discarded**.



Model-Based vs. Instance-Based Learning

Instance-based

- ▶ Instance-based learning algorithms use the **whole dataset as the model**.
- ▶ In instance-based learning, the **algorithm doesn't explicitly learn a model but instead memorizes the training data**.
- ▶ Predictions for new data are made by **comparing them directly to the stored instances**.



Shallow vs. Deep Learning

- ▶ A shallow **learning algorithm** learns the parameters of the **model** directly from the features of the **training examples**.
- ▶ Most machine learning algorithms are **shallow**.
- ▶ specifically those that build **neural networks with more than one layer between** input and output. **Such neural networks are called deep neural networks**. In deep neural network learning (or, simply, deep learning),

Application of ML

- ▶ Image Recognition
- ▶ Speech Recognition
- ▶ Recommender Systems
- ▶ Fraud Detection
- ▶ Self Driving Cars
- ▶ Medical Diagnosis
- ▶ Stock Market Trading

Main Challenges of Machine Learning

Insufficient Quantity of Training Data

- ▶ it takes a lot of data for **most Machine Learning algorithms** to work properly.
- ▶ Even for very simple problems you typically need **thousands of examples**, and for complex problems such as image or speech recognition you may need **millions of examples**.

Non representative Training Data

- ▶ In order **to generalize well**, it is crucial that your training data be representative of the **new cases** you want to **generalize** to. This is true whether you use **instance-based** learning or model-based learning.

Poor-Quality Data

- ▶ Obviously, if your training data is **full of errors, outliers, and noise** (e.g., due to poor quality measurements), it will **make it harder for the system to detect the underlying patterns**, so your system is less likely to **perform well**.

Main Challenges of Machine Learning

Irrelevant Features

- ▶ As the saying goes: **garbage in, garbage out**. Your system will only be capable of learning if the training data contains **enough relevant features and not too many irrelevant ones**.
- ▶ A critical part of the success of a Machine Learning project is **coming up with a good set of features to train on**.

Overfitting the Training Data

- ▶ It means that the model performs well on **the training data, but it does not generalize well**. Not **learn generalized**.

Underfitting the Training Data

- ▶ Underfitting is the opposite of overfitting: it occurs when your model is too **simple to learn the underlying structure of the data**.

Bias and Variance in Machine Learning

Bias

- ▶ Bias is one type of **error that occurs due to wrong assumptions** about data such as assuming data is **linear when in reality, data follows a complex function**.
- ▶ Bias is a systematic error that occurs due to **wrong assumptions** in the machine learning process.
- ▶ **Low Bias**
 - ➔ In this case, the model will closely **match the training dataset**.
- ▶ **High Bias**
 - ➔ the model will **not match the training dataset closely**.
- ▶ For example, a **linear regression model** may have a **high bias** if the data has a **non-linear relationship**.
- ▶ The high-bias model will not **be able to capture the dataset trend**. It is considered as the **underfitting model which has a high error rate**. It is due to a very **simplified algorithm**.

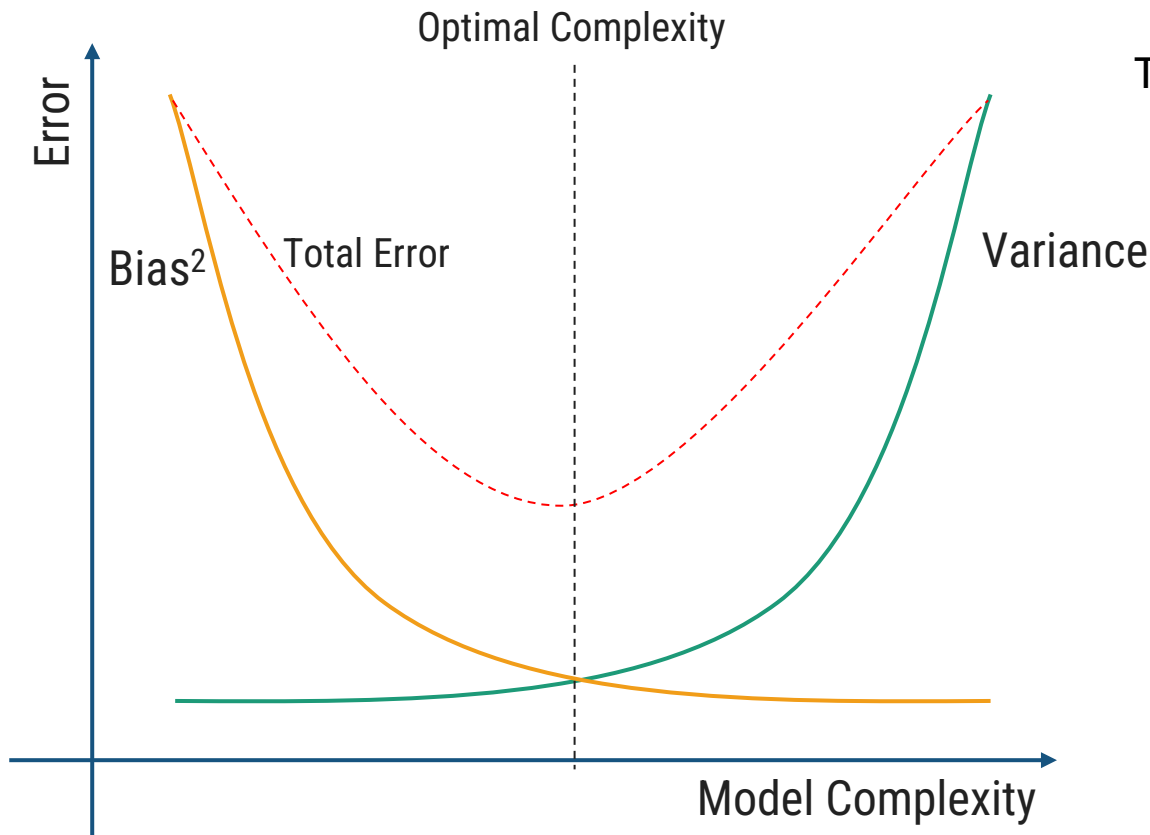
Bias and Variance in Machine Learning

► Variance

- Variance is the **variability of the model** that how much it is **sensitive to another subset of the training dataset**.
- In machine learning **variance is the amount by which the performance of a predictive model changes when it is trained on different subsets of the training data**.
- how much it can adjust on the **new subset of the training dataset**.
- **Low variance:**
 - Low variance means that the model is **less sensitive to changes in the training data** and can produce consistent estimates of the target function with **different subsets of data from the same distribution**
- **High variance:**
 - High variance **means that the model is very sensitive** to changes in the training data and can result in significant changes in the estimate of **the target function when trained on different subsets** of data from the same distribution.

Bias-Variance Trade-Off

- ▶ The goal is to achieve a **balance where the model generalizes well** on unseen data.
- ▶ This involves **minimizing the combined error from bias and variance**.



$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

- ▶ Increasing model complexity typically decreases bias but increases variance.
- ▶ Decreasing model complexity typically increases bias but decreases variance.
- ▶ **High Bias and Low Variance - Underfitting**
- ▶ **Low Bias and High Variance - Overfitting**

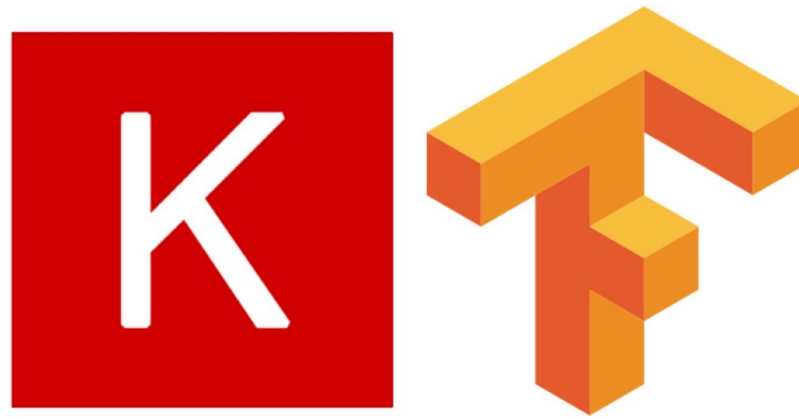
Tools used for Machine Learning : Scikit-learn

- ▶ The Scikit-learn library is one of many Scikit libraries that build on the capabilities provided by NumPy and SciPy to allow Python developers to perform domain specific tasks.
- ▶ Scikit-learn library focuses on machine Learning and data analysis, it provides access to following sort of functionality:
 - Classification
 - Regression
 - Clustering
 - Dimensionality reduction
 - Model selection
 - Pre-processing
- ▶ Scikit-learn is the most important library we are going to learn in this subject



Tools used for Machine Learning : Keras and TensorFlow

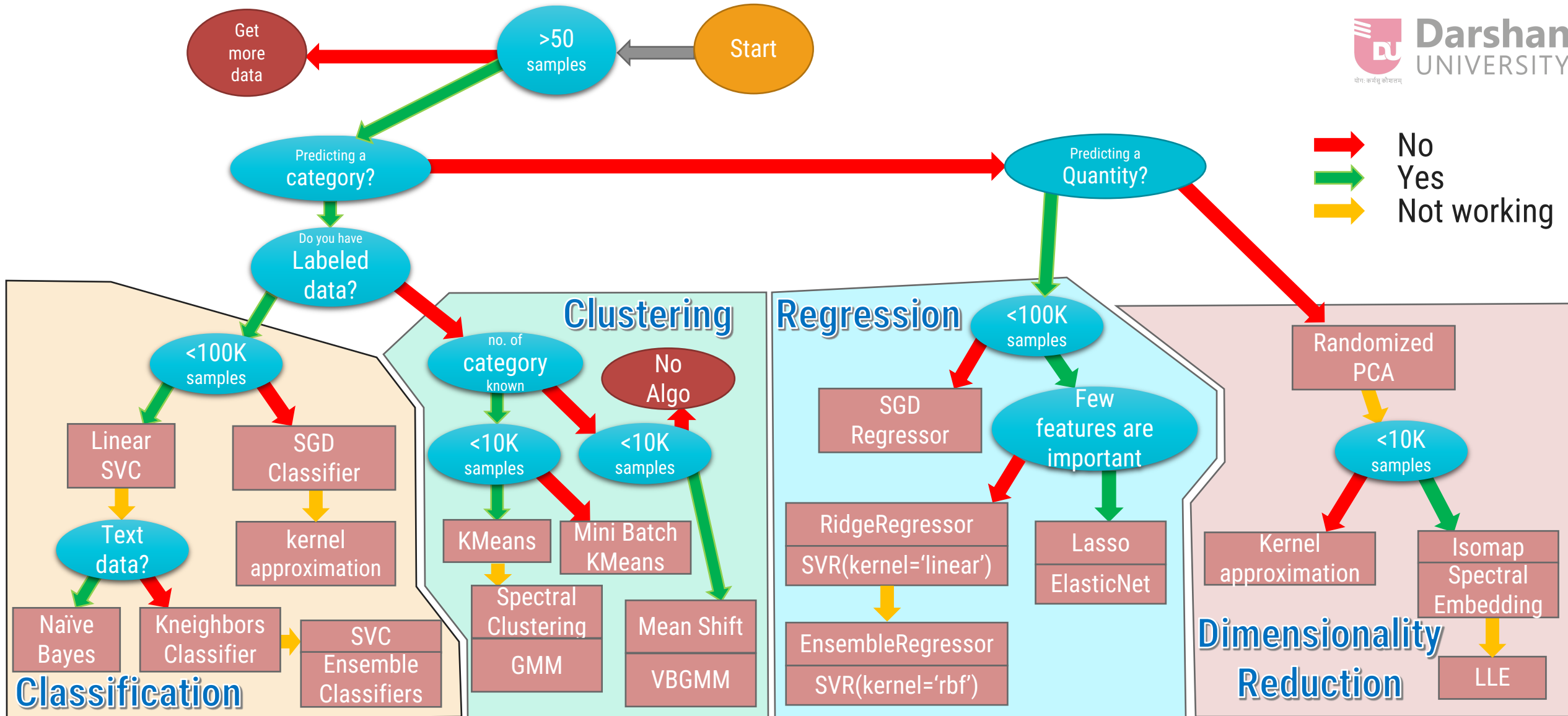
- ▶ Keras is an application programming interface (API) that is used to train deep learning models.
- ▶ An API often specifies a model for doing something, but it doesn't provide an implementation.
- ▶ TensorFlow is an implementation for the keras, there are many other implementations for the keras like
 - ➔ Microsoft's cognitive Toolkit, CNKT
 - ➔ Theano



Classes in Scikit-learn

- ▶ Understanding how classes work is an important prerequisite for being able to use the **Scikit-learn package appropriately**.
- ▶ Scikit-learn is the package for machine learning and data science experimentation **favored by most data scientists**.
- ▶ It contains wide range of well-established learning algorithms, error functions and testing procedures.
- ▶ Install: `conda install scikit-learn`
- ▶ There are four class types covering all the basic machine-learning functionalities,
 - **Classifying**
 - **Regressing**
 - **Grouping by clusters**
 - **Transforming data**

Choosing a correct algorithm in Scikit-learn



Basic Structure of Scikit-Learn

Step – 01 (import)

```
from sklearn.family import Model
```

Example

```
from sklearn.linear_model import LinearRegression
```

Step – 02 (create object)

```
m = Model(parameters)
```

Example

```
lr = LinearRegression()
```

Step – 03 (split data) (Only For Supervised Learning)

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.3)
```

Step – 04 (Train Model)

```
m.fit(X_train,y_train)    #For Supervised Learning
```

OR

```
m.fit(X_train)            #For Unsupervised Learning
```

Step – 05 (Predict) (Only For Supervised Learning)

```
predictions = m.predict(X_test)
```

Step – 06 (Evaluate Model)

- Evaluate the model and update the parameters if required.
- Evaluating technique will be different for regression, classification and clustering.
- We will explore all the techniques in later sessions.

Example Model Evaluation for Regression

```
from sklearn import metrics
```

```
print(metrics.mean_squared_error(ytest,predictions))
```

train_test_split()

- ▶ Split arrays or **matrices** into random train and test subsets.
- ▶ The output of the train_test_split function is **X_train, X_test, y_train, and y_test**, which are all NumPy arrays. X_train contains **75% of the rows of the dataset**, and X_test contains the **remaining 25%**
- ▶ Before making the split, the train_test_split function **shuffles the dataset** using a **pseudorandom** number generator.
- ▶ we provide the pseudorandom number generator with a fixed seed using the random_state parameter. This will make the outcome deterministic, so this line will always have the same outcome.

Demo.py

```
1 from sklearn.model_selection import train_test_split
2 X_train, X_test, y_train, y_test = train_test_split(iris_dataset['data'],
  iris_dataset['target'], random_state=0)
```